Experiment No. 11

Title

To Develop and Deploy a Milk Dairy Management System Using React.js, Node.js, Express.js, and MongoDB.

Objectives

1. To develop a web application for managing milk dairy records.
2. To store customer and milk collection details in MongoDB.
3. To perform basic operations such as adding and viewing milk records.
4. To understand frontend–backend integration and cloud deployment.

Theory

A Milk Dairy Management System is a web application used to manage daily milk collection, customer details, milk quantity, milk rates, and payment records.

- **React.js:** Creates the user interface.
- **Node.js:** Runs the backend application.
- **Express.js:** Handles API requests.
- **MongoDB:** Stores customer and milk collection records.
- **Cloud Hosting:** Makes the application accessible online.

Problem Statement

Develop a Milk Dairy Management System that allows dairy staff to enter customer names, milk types, milk quantities, and milk rates. The system should calculate the total milk amount and store the records in MongoDB.

Main Features

1. Add customer details.
2. Record daily milk collection.
3. Select cow milk or buffalo milk.
4. Calculate the total amount.
5. Display milk collection records.
6. Store data in MongoDB.
7. Deploy the application on a cloud platform.

Sample Calculation

Total Amount = Milk Quantity × Rate per Litre

Example:

Milk Quantity = 10 litres

Rate = 60 per litre

Total Amount = 10 × 60 = 600

Result

A Milk Dairy Management System can be developed using React.js, Node.js, Express.js, and MongoDB to manage daily milk collection and payment records.

Part of the project	Where to host it
React.js frontend	Vercel or Render
Node.js + Express.js backend	Render
MongoDB database	MongoDB Atlas